

EmpathyAI

Project Final Report

LLM Fine-tuning for Empathy Level Classification
OPELA Dataset based GPT-4.1 Nano & DeepSeek 7B Optimization

Base Model Accuracy:	15.70%
GPT Fine-tuned Accuracy:	33.49%
DeepSeek QLoRA Accuracy:	49.19%
Best Improvement:	+33.49%p

2025-12-17

Based on Smilegate AI & Seoul National University Research Data

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1. Introduction

1.1 Research Background

Empathy is a core element of user experience in conversational AI systems. This project developed a system that automatically classifies empathy levels in AI (persona) responses during Korean conversations. We fine-tuned two models using the OPELA (Open-domain conversations by Personas with Empathy, Long-term memory, and Attractive personality) dataset:

- GPT-4.1 Nano: Fine-tuned via OpenAI API
- DeepSeek 7B: Fine-tuned via QLoRA on Google Colab

1.2 Research Objectives

- Automatic empathy level classification in Korean persona-user dialogues
- Comparison of API-based vs. open-source model fine-tuning approaches
- Evaluation of QLoRA efficiency for large language model training

Table 1. Empathy Level Definitions

Level	Label	Description
0	Not Applicable	Situations where empathy does not apply
1	Empathy Failure	Failed empathy (ignored or inappropriate)
2	Low Empathy	Low level empathy (minimal response)
3	Moderate Empathy	Moderate empathy (appropriate response)
4	High Active Empathy	High active empathy (deep understanding)

Note. Empathy levels were labeled by third-party evaluators using majority voting.

2. Dataset

2.1 OPELA Dataset Overview

The OPELA dataset was collected through a joint research project by Smilegate AI and Seoul National University. It consists of actual persona-user role-play conversations between crowdworkers, covering various daily topics with 15 to 80 turns per conversation.

2.2 Data Statistics

Table 2. Descriptive Statistics of the OPELA Dataset

Statistic	Value
Total Samples	14,825
Unique Conversations	533
Avg Turns/Conversation	30.14
Avg User Text Length	48.0 chars
Avg Persona Text Length	45.5 chars

Note. Text length is measured in Korean characters.

2.3 Label Distribution

Table 3. Empathy Label Distribution in the Full Dataset

Label	Count	Percentage	Cumulative
0	4,167	28.1%	28.1%
1	660	4.5%	32.6%
2	1,806	12.2%	44.7%
3	7,198	48.6%	93.3%
4	994	6.7%	100.0%
Total	14,825	100.0%	-

Note. Labels 0 (Not Applicable) and 3 (Moderate Empathy) have the highest proportions.

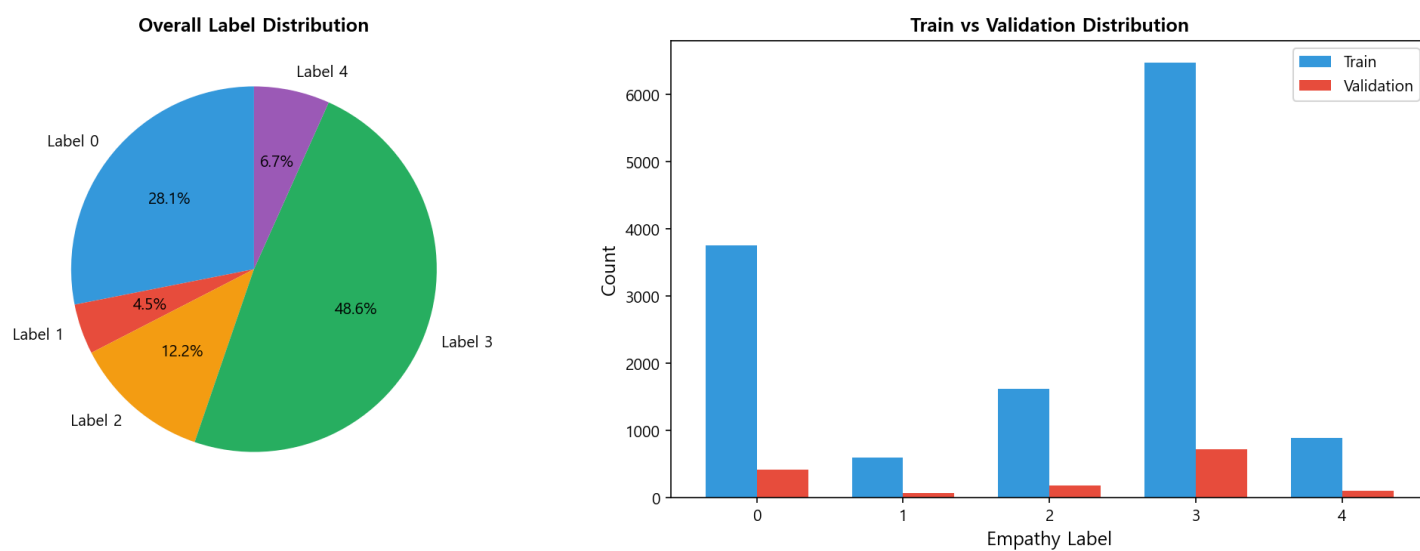


Figure 1. Empathy Label Distribution

Note. Left: Overall dataset distribution. Right: Train vs Validation distribution comparison.

2.4 Train/Validation Split

Table 4. Train and Validation Set Label Distribution

Label	Train	Train %	Val	Val %
0	3,750	28.1%	417	28.1%
1	594	4.5%	66	4.4%
2	1,625	12.2%	181	12.2%
3	6,478	48.6%	720	48.5%
4	894	6.7%	100	6.7%
Total	13,341	100%	1,484	100%

Note. Stratified sampling was used with a 90:10 split ratio.

3. Methodology

3.1 Model Configuration

Table 5. Model Configuration Comparison

Parameter	GPT-4.1 Nano	DeepSeek 7B
Base Model	gpt-4.1-nano	DeepSeek-R1-Distill-Qwen-7B
Parameters	Unknown (API)	7 Billion
Training Method	SFT (API)	QLoRA (4-bit + LoRA)
Platform	OpenAI API	Google Colab (H100)
LoRA Rank	N/A	16
Learning Rate	Auto	2e-4
Epochs	3	3
Training Time	~30 min	~3 hours

Note. GPT fine-tuning via OpenAI API; DeepSeek via QLoRA on Colab.

3.2 QLoRA Methodology

QLoRA (Quantized Low-Rank Adaptation) enables efficient fine-tuning of large models:

1. 4-bit Quantization: Reduces model weights to 4-bit precision (NF4)
2. LoRA Adapters: Trains only 0.5% of parameters while freezing the rest
3. Double Quantization: Further reduces memory by quantizing quantization constants

This approach reduced GPU memory requirement from 28GB to 6GB for the 7B model.

3.3 Prompt Design

The prompt structure used for model training and inference:

[System] You are an empathy classifier for Korean persona-user dialogues. Output JSON with "empathy_label" (0-4).

[User] Classify the empathy level of the PERSONA's reply.

USER: [user utterance]

PERSONA: [persona response]

Return JSON only.

4. Experimental Results

4.1 Overall Performance Comparison

Table 6. Overall Model Performance Comparison

Metric	GPT Base	GPT FT	DeepSeek	Best Impr.
Accuracy	15.70%	33.49%	49.19%	+33.49%p
Macro Precision	0.2272	0.2270	0.3067	+0.0795
Macro Recall	0.1939	0.2326	0.2284	+0.0345
Macro F1	0.1452	0.2275	0.2034	+0.0823
Correct/Total	233/1484	497/1484	730/1484	+497

Note. DeepSeek achieved highest accuracy (49.19%); GPT FT achieved highest Macro F1 (0.2275).

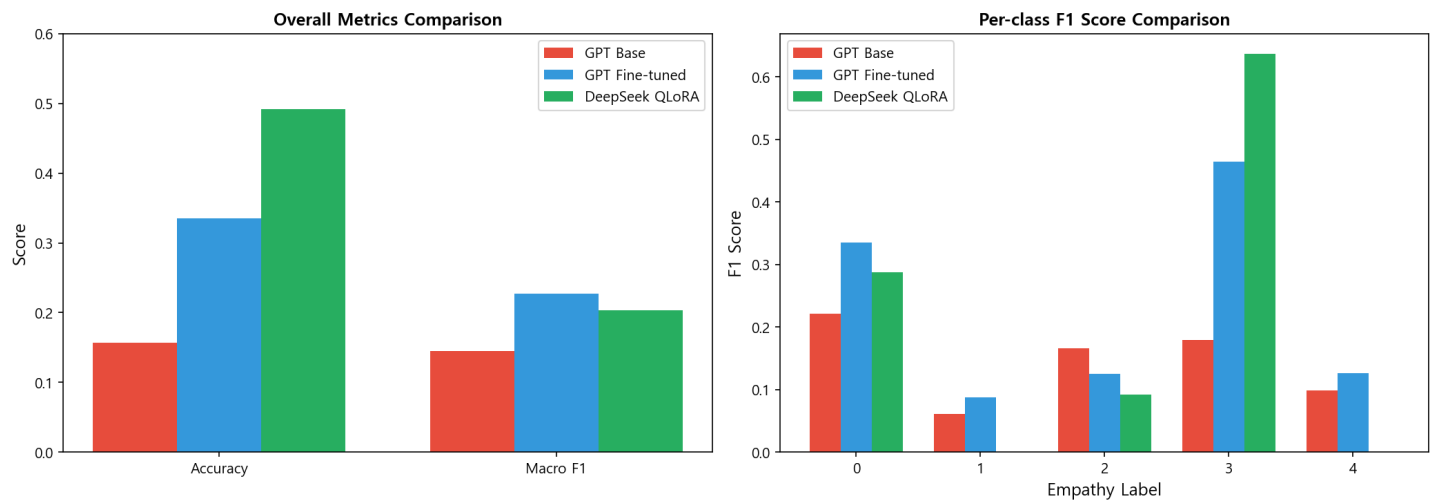


Figure 2. Model Performance Comparison

Note. Left: Overall metrics comparison. Right: Per-class F1 score comparison.

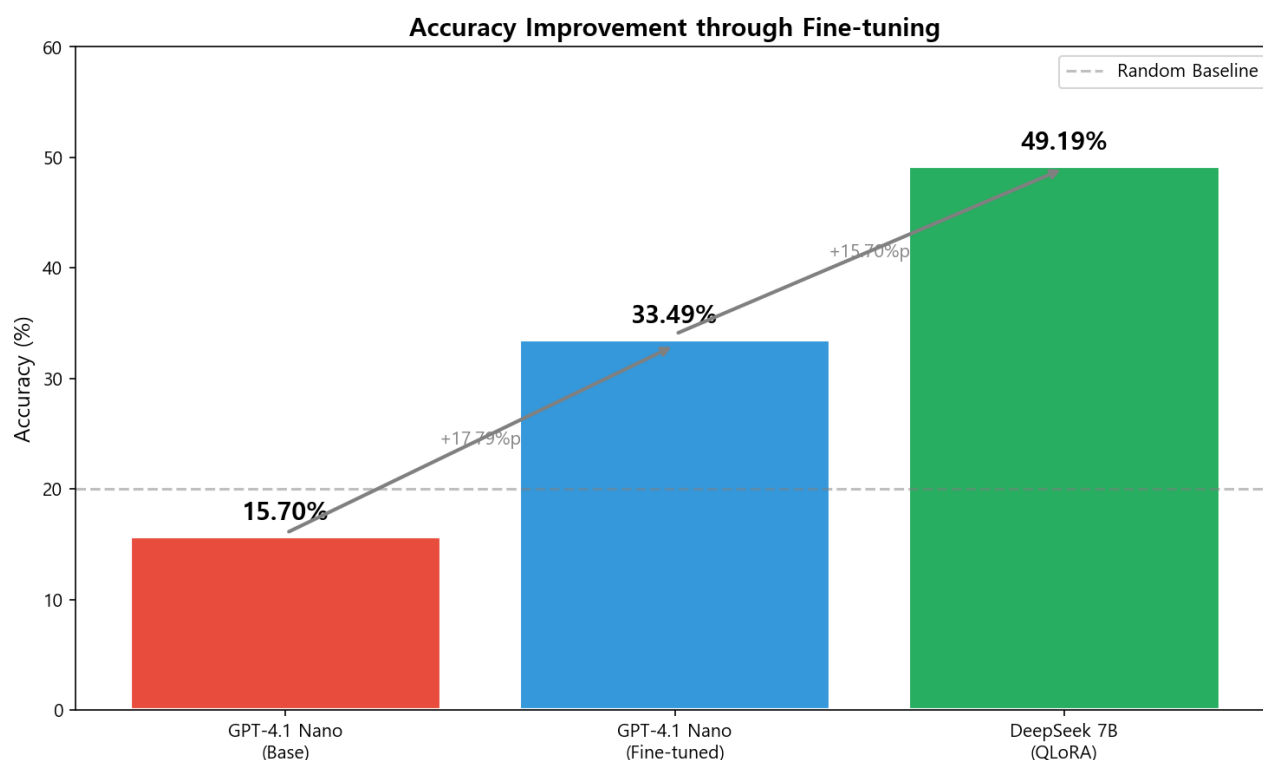


Figure 3. Accuracy Improvement through Fine-tuning

Note. DeepSeek QLoRA achieved +33.49%p improvement over base model.

4.2 Per-Class Performance Analysis

Table 7. Per-Class Performance Metrics

Label	Base F1	GPT FT F1	DeepSeek F1
0	0.221	0.335	0.288
1	0.061	0.087	0.000
2	0.166	0.125	0.092
3	0.179	0.464	0.637
4	0.099	0.126	0.000

Note. DeepSeek excels at Label 3 (F1=0.637) but fails on Labels 1, 4.

4.3 Confusion Matrix Analysis

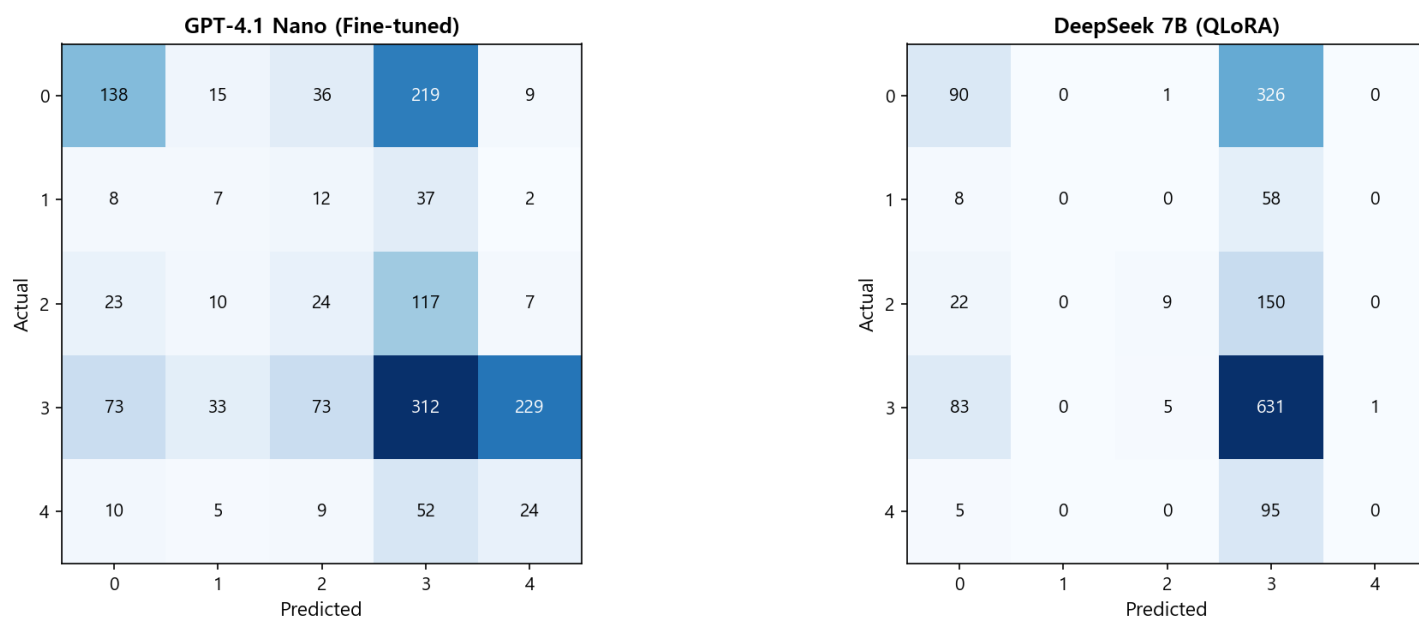


Figure 4. Confusion Matrices for Fine-tuned Models

Note. Left: GPT Fine-tuned. Right: DeepSeek QLoRA. DeepSeek shows strong bias toward Label 3.

5. Text Analysis

5.1 Text Length Distribution

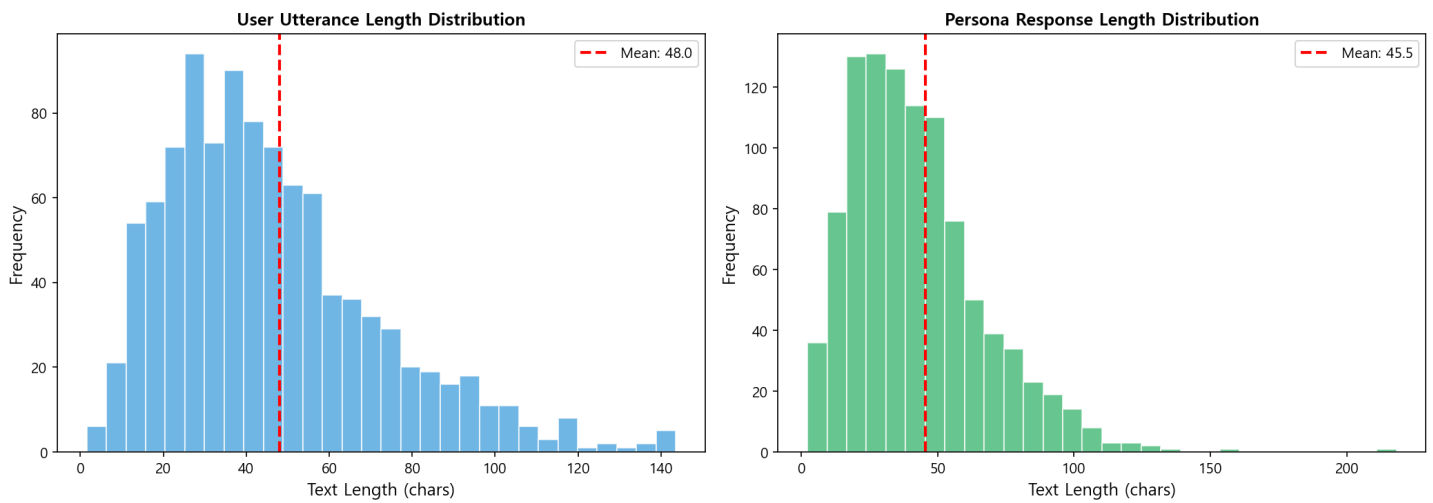


Figure 5. Text Length Distribution

Note. Left: User utterance length. Right: Persona response length. Red dashed line indicates mean.

5.2 Response Length by Empathy Level

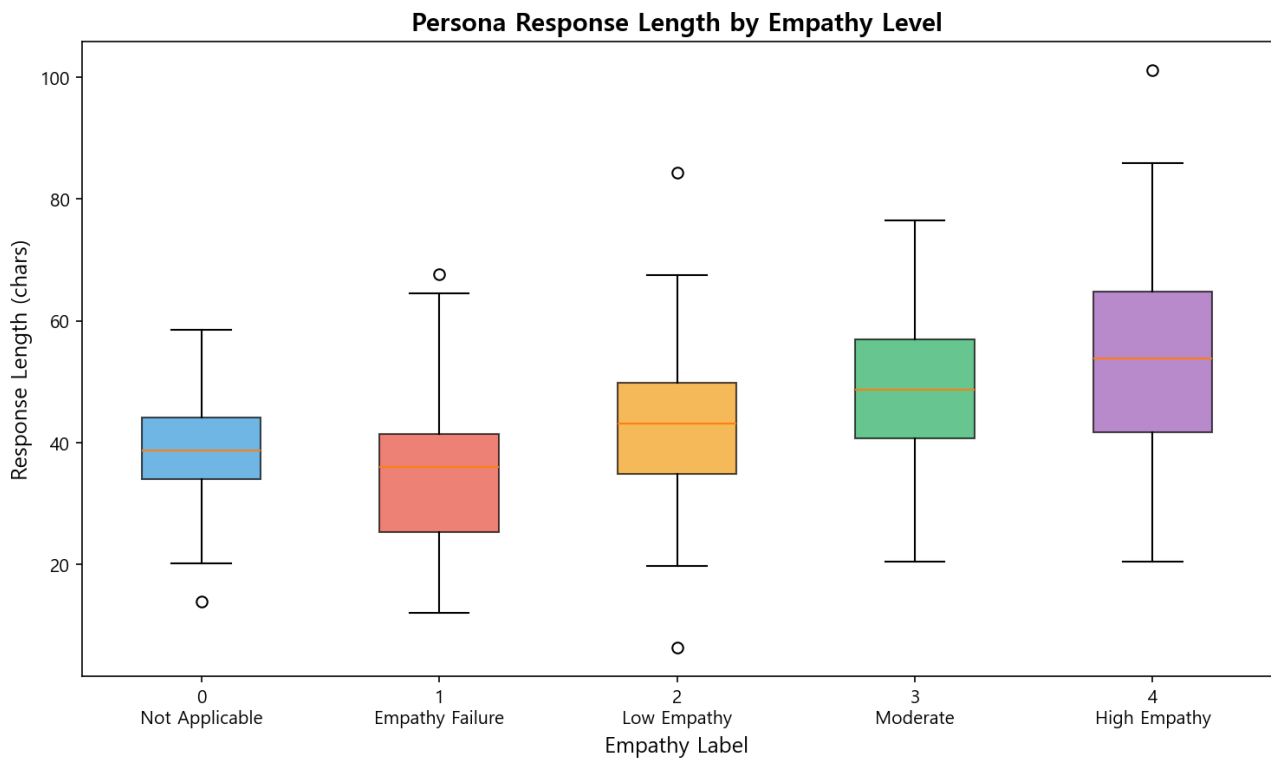


Figure 6. Persona Response Length by Empathy Level

Note. Higher empathy levels tend to have longer responses.

6. Conclusion

6.1 Key Achievements

This project developed a system that automatically classifies empathy levels in Korean conversations using the OPELA dataset. Key achievements include:

- Built empathy classification model using OPELA dataset (14,825 samples)
- GPT-4.1 Nano fine-tuning: 15.70% -> 33.49% (+17.79%p)
- DeepSeek 7B QLoRA fine-tuning: 15.70% -> 49.19% (+33.49%p)
- Demonstrated QLoRA efficiency: 7B model trained on free Colab GPU
- Identified trade-off between accuracy and class balance

6.2 Model Comparison Summary

Table 8. Final Model Comparison

Aspect	GPT-4.1 Nano FT	DeepSeek QLoRA
Accuracy	33.49%	49.19%
Macro F1	0.2275	0.2034
Minority Class	Better	Poor
Training Cost	API credits	Free (Colab)
Reproducibility	Limited	Full

Note. Choice depends on priority: accuracy (DeepSeek) vs. class balance (GPT).

6.3 Limitations and Future Work

Limitations:

- Severe class imbalance (Label 3 = 48.6% of data)
- DeepSeek ignores minority classes (Labels 1, 4)
- 5-class classification may be too fine-grained

Future Work:

1. Class simplification: 5-class -> 3-class (Low/Medium/High)
2. Data augmentation for minority classes
3. Ensemble of GPT and DeepSeek models
4. LLaMA 3.1 8B fine-tuning for comparison
5. Class-weighted loss function for balanced training

References

Smilegate AI & Seoul National University (2022). OPELA: Open-domain conversations by Personas with Empathy, Long-term memory, and Attractive personality. GitHub: <https://github.com/smilegate-ai/OPELA>

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